20. **Assertion (A) :** Lines given by $x = py + q$, $z = ry + s$ and $x = p'y + q'$, $z = r'y + s'$ are perpendicular to each other when $pp' + rr' = 1$. **1**

Reason (R) : Two lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$ are perpendicular to each other if $\vec{b}_1 \cdot \vec{b}_2 = 0$.

SECTION – B

This section comprises Very Short Answer (VSA) type questions of **2** marks each.

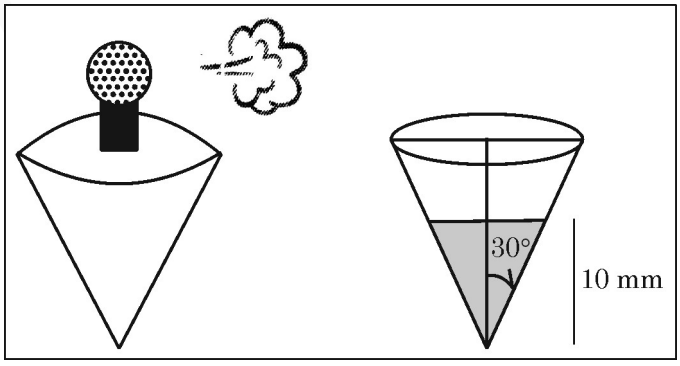
21. (a) Check whether function f(x) defined as **2**

$$f(x) = \begin{cases} \dfrac{|x-3|}{2(x-3)}, & x < 3 \\ \dfrac{x-6}{6}, & x \geq 3 \end{cases}$$ is continuous at $x = 3$ or not ?

OR

(b) If $\sqrt{3}(x^2 + y^2) = 4xy$, then find $\dfrac{dy}{dx}$ at $\left(\dfrac{1}{2}, \dfrac{\sqrt{3}}{2}\right)$. **2**

22. A room freshner bottle in the shape of an inverted cone sprays the perfume at regular intervals such that volume of the perfume in the bottle decreases at the steady rate of 1 mm^3/min. Find the rate at which level of perfume is dropping at an instant when level of perfume in the bottle is 10 mm, if the semi-vertical angle of conical bottle is $\dfrac{\pi}{6}$. **2**



33. (a) If $A = \begin{bmatrix} 0 & 2 & 1 \\ -2 & -1 & -2 \\ 1 & -1 & 0 \end{bmatrix}$, find A^{-1} and use it to solve the following system of equations :

$-2y + z = 7$, $2x - y - z = 8$, $x - 2y = 10$ **5**

OR

(b) If $\begin{bmatrix} 3 & -1 & \sin 3x \\ -7 & 4 & \cos 2x \\ -11 & 7 & 2 \end{bmatrix}$ is a singular matrix, then find all values of x where $x \in \left[0, \frac{\pi}{2}\right]$. **5**

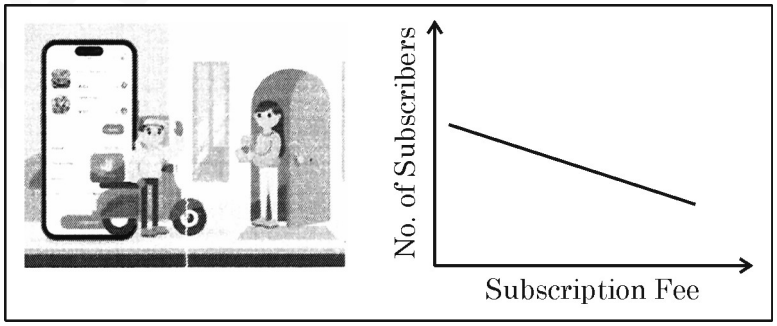
34. If $x = \cos t$, $y = \cos mt$, prove that $(1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + m^2y = 0$. **5**

35. Check whether the lines given by $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{5} = \frac{y-1}{2} = z$ are parallel or not. If parallel, find the distance between them, otherwise find their point of intersection, if the lines are intersecting. **5**

SECTION – E

This section comprises of **3** case study based questions of **4** marks each.

36. An online delivery company in a city has 5000 subscribers and collects annual subscription fees of ₹ 300 per subscriber for unlimited free deliveries.



The company wishes to increase the annual subscription fee. It is predicted that, for every increase of ₹ 1, ten subscribers will discontinue. Assume that the company increased the annual fee by ₹ x.

30. Evaluate : $\int_0^1 x \sin^{-1}x \, dx$ **3**

31. (a) Find $\int \sqrt{\frac{x+2}{x-2}} \, dx$ **3**

OR

(b) Find : $\int \frac{x^2}{(x^2+9)(x^2+16)} \, dx$ **3**

SECTION – D

This section comprises Long Answer (LA) type questions of **5** marks each.

32. If $x = 3 \sin t - \sin 3t$, $y = 3 \cos t - \cos 3t$ **5**

find $\frac{dy}{dx}$ and prove that $\frac{d^2y}{dx^2} = \frac{-\text{cosec}^3 2t \cdot \text{cosec } t}{3}$

33. Prove that the line through points A(0, –1, –1) and B(4, 5, 1) intersects the line through points C(3, 9, 4) and D(–4, 4, 4). Hence, write the equation of line passing through the point of intersection of lines AB and CD as well as origin. **5**

34. (a) A relation R is defined on Z, the set of integers, as **5**

$R = \{(x, y) : |x - y| \text{ is divisible by a prime number 'p', } x, y \in Z\}$

check whether R is an equivalence relation or not.

OR

(b) A function $f : R - \left\{\frac{3}{5}\right\} \longrightarrow R - \left\{\frac{3}{5}\right\}$ is defined as $f(x) = \frac{3x+2}{5x-3}$. Show that f is one-one and onto. **5**

	For Visually Impaired: Find the range of $\frac{\pi}{2} + \sec^{-1}\left(\frac{x}{2}\right)$ (A) $[0, \pi]$ (B) $[0, \pi] - \left\{\frac{\pi}{2}\right\}$ (C) $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ (D) $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right] - \{\pi\}$	
Q2.	Domain of the function: $\sec^{-1}(2x)$ is (A) $(-\infty, \infty)$ (B) $\left(-\frac{1}{2}, \frac{1}{2}\right)$ (C) $\left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$ (D) $\left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$	**1**
Q3.	Value of $\cot^{-1}\left(2\cos\left(\frac{1}{2}\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)\right)$ (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$	**1**
Q4.	If $A = \begin{bmatrix} 2 & 1 & -2 \\ 3a & 0 & 4 \\ b & 2c & 1 \end{bmatrix}$ is a symmetric matrix, then value of $\sqrt{6a + 2b + 3c}$ is: (A) -2 (B) 2 (C) ± 2 (D) 4	**1**
Q5.	If P and Q are matrices of same order such that $P = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, $\lvert Q \rvert = 3$, then find $\lvert adj(PQ) \rvert$ (A) -3 (B) 1 (C) 6 (D) 36	**1**
Q6.	If $\begin{bmatrix} 1 & x \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \end{bmatrix} = O$, then value of x is (A) -2 (B) -1 (C) 1 (D) 2	**1**
Q7.	The derivative of $\cos^{-1}(2x^2 - 1)$ with respect to $\cos^{-1}(x)$ is (A) 2 (B) ± 1 (C) $\frac{1}{2}$ (D) -3	**1**
Q8.	If a, b, c are arbitrary constants such that the function defined by $f(x) = \begin{cases} ax^2 + b, & b \neq 0, \ x \leq 1 \\ bx^2 + ax + c, & x > 1 \end{cases}$ is differentiable at $x = 1$, then which of the following could be a triplet for (a, b, c)? (A) (3,2,0) (B) (0,0,3) (C) $\left(\frac{3}{2}, 2, 0\right)$ (D) $\left(3, \frac{3}{2}, 0\right)$	**1**
Q9.	If $y = a\left\{x + 2a \sin\left(\frac{x}{a}\right)\right\}$, $a > 0, x \in [0, \pi a]$; is a decreasing function, then the interval in which x lies is (A) $\left(\frac{5\pi}{6}, \pi\right]$ (B) $\left(\frac{2\pi}{3}, \pi\right]$ (C) $\left(\frac{5\pi a}{-6}, \pi a\right]$ (D) $\left(\frac{2\pi a}{3}, \pi a\right]$	**1**

*There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27

	Assertion-Reason Based Questions **Directions: Question numbers 19 and 20 are Assertion and Reason based questions carrying 1 mark each. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R).** **Select the correct answer from the options (A), (B), (C) and (D) as given below:** (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.	
Q19.	Assertion: The value of the integral $\int_{-1}^{1}\|x-2\|\,dx$ is 4 Reason: If $f(-x) = f(x)$ then $\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx$	**1**
Q20.	Assertion: The general solution of a differential equation represents a family of curves. Reason: The derivative of an arbitrary constant is zero.	**1**
SECTION – B (5 x 2 = 10) **This section comprises of 5 very short answer (VSA) type questions of 2 marks each**		
Q21 (A).	Evaluate $\int \frac{1}{x\sqrt{x^6-1}}dx$	**2**
	OR	
Q21 (B).	Evaluate $\int \frac{\cos x - \sin x}{\sqrt{\sin(2x)}}\,dx$	
Q22.	If $y = \sin(x+y)$ and $\frac{dy}{dx} = 0$, then prove that $y = \pm 1$.	**2**
Q23 (A).	A line passing through the points $(2,6,5), and\ (3,p,7)$ is parallel to the line joining the points $(-4,3,4)$ and $(-1,0,q)$. Find the value of $\frac{q}{p}$.	**2**
	OR	
Q23 (B).	A rectangle ABCD is such that $\overrightarrow{AC} = \vec{u}, and\ \overrightarrow{BD} = \vec{v}$, then find the vectors representing four sides of the rectangle (in a cyclic order starting from $\overrightarrow{AB}$).	
Q24.	For two non-zero vectors $\vec{a}$ and $\vec{b}$, $\|\vec{a}\| = \sqrt{6}$, $\vec{a}\cdot\vec{b} = 1$ and $\vec{b}\times\vec{a} = 3\hat{\imath} + 4\hat{\jmath} - 2\hat{k}$. Find $\|\vec{b}\|$.	**2**

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